SMART STUDENT ATTENDANCE AND MONITORING SYSTEM

An intuitive system for automated attendance recording, monitoring and report generation

ACCURATE — Error free records
FAST — Save time
TRANSPARENT — Real-time visibility
SECURE — Role based access

GOAL DIRECTED DESIGN PROCESS

1. RESEARCH
   Interviews with faculty, students
   Observe current process
   Identify pain points
2. MODELING
   Create personas
   Understand goals, needs, behaviors
3. REQUIREMENTS DEFINITION
   User goals to requirements
   Tasks, scenarios, use cases
4. FRAMEWORK DEFINITION
   Information architecture
   Task flows, navigation
   Wireframes
5. REFINEMENT
   High fidelity UI design
   Apply HMI principles
6. VALIDATION
   Usability testing
   Analyze & refine
7. FINAL DESIGN / IMPLEMENTATION
   Final prototype
   Documentation

USER PERSONAS



Prof. Meera — Faculty
Goal: Mark attendance quickly and accurately
Pain Point: Manual process takes time



Rohan — Student
Goal: Track my attendance and shortage
Pain Point: Don't know real-time status

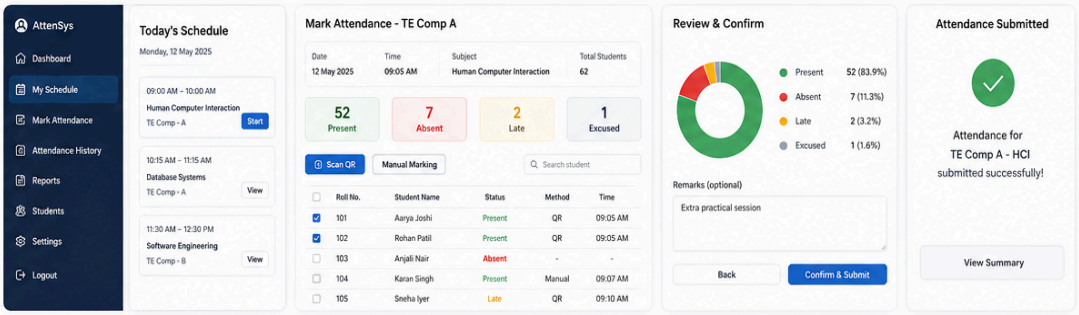


Dr. Sharma — Administrator
Goal: Monitor attendance across all classes
Pain Point: Hard to get consolidated reports

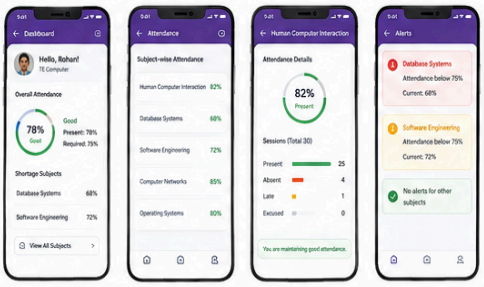
FACULTY – ATTENDANCE WORKFLOW

Login → Today's Schedule → Start Attendance → Verify Students → Review & Correct → Submit → Summary

FACULTY INTERFACE



STUDENT INTERFACE



ADMINISTRATOR DASHBOARD



KEY FEATURES

- Automated Attendance — QR / Face / ID Based
- Real-time Monitoring — Instant visibility
- Attendance Alerts — For students & faculty
- Reports & Analytics — Subject, class, faculty wise
- Secure & Role Based — Data privacy & security
- Easy Correction — Legitimate changes

HMI PRINCIPLES APPLIED

- Visibility of System Status — Clear counts, color coding, progress
- Recognition rather than Recall — Icons, labels, search, lists
- Consistency & Standards — Same layout, colors, patterns
- Error Prevention & Recovery — Confirm before submit, validation
- Feedback — Instant messages, alerts, progress
- Flexibility & Efficiency — Multiple ways to perform tasks
- Accessibility — Readable text, contrast, icons

STATUS COLORS

- Present
- Absent
- Late
- Excused

Goal-Directed Design Process – Smart Student Attendance and Monitoring System

GDD Phase	Application to the Attendance System	Expected Student Output
1. Research	Identify the current attendance process and understand how faculty, , and administrators record, verify, and monitor attendance. Observe problems such as manual entry, proxy attendance, calculation errors, and difficulty in finding attendance records.	User interviews, observations, questionnaires, existing-system study
2. Modeling	Develop user models/personas based on research findings. Typical users are **Faculty**, **Student**, and **Department Administrator**. Identify their goals, behaviors, frustrations, and technology usage.	Personas, user goals, behavioral patterns
3. Requirements Definition	Convert user goals into functional and interaction requirements. For example, faculty should be able to quickly start an attendance session, verify , correct legitimate errors, and submit attendance. should be able to view their attendance percentage and alerts.	User requirements, task analysis, user scenarios, use cases
4. Framework Definition	Design the overall interaction framework: navigation, dashboard structure, attendance workflow, information hierarchy, feedback mechanisms, error handling, and screen relationships.	Information architecture, task flow, navigation map, wireframes
5. Refinement	Develop detailed interface designs. Apply HMI principles such as visibility, feedback, consistency, affordance, error prevention, recognition rather than recall, and accessibility.	High-fidelity prototype, interactive prototype
6. Validation	Conduct usability testing with representative users. Ask faculty/ to perform tasks such as marking attendance, correcting an entry, checking attendance percentage, and generating a report. Identify usability problems and refine the design.	Usability test plan, test results, observations, design improvements

7. Final Design / Implementation	Develop the final interactive prototype or application based on validated design decisions. Demonstrate how the system reduces faculty effort and improves attendance visibility for and administrators.	Final prototype/application and design documentation

1. Identify the Users

Primary users

- **Faculty** – records and manages attendance.
- – view their attendance and attendance alerts.
- **Department Administrator** – monitors attendance across classes/divisions and generates reports.

2. Identify User Goals

User	Primary Goal	Secondary Goals
Faculty	Record attendance quickly and accurately	Correct errors, view attendance history, generate reports
Student	Know current attendance status	View subject-wise attendance, receive shortage alerts
Administrator	Monitor attendance across classes	Generate reports, identify attendance shortages, track faculty compliance

3. Example User Scenario

Scenario: Faculty taking attendance

A faculty member enters the classroom and opens the attendance system. The system automatically identifies the scheduled lecture and displays the corresponding class and subject. The faculty initiates the attendance session and are automatically identified through the selected authentication mechanism. The interface clearly displays attendance status and highlights exceptions. Before submission, the faculty reviews the attendance summary and confirms the record. can subsequently view their updated attendance percentage.

4. Goal-Directed User Flow

Faculty

Login → Today's Schedule → Select Lecture → Start Attendance → Student Verification → Review Attendance → Correct Exceptions → Confirm & Submit → Attendance Summary

Student

Login → Dashboard → Attendance → Subject-wise Attendance → Percentage → Shortage/Alert → Attendance History

Administrator

Login → Attendance Dashboard → Select Division/Subject → View Statistics → Identify Shortage → Generate Report

5. Important HMI Problems Should Solve

Instead of simply creating an attendance database, the mini-project should investigate:

- How can attendance be recorded with **minimum interaction time**?
- How can the interface prevent **accidental or duplicate attendance entries**?
- How can faculty immediately recognize **absent/present/unverified** ?
- How can the system provide clear **feedback after attendance submission**?
- How can understand their attendance status **without interpreting complex tables**?
- How should the system highlight **attendance shortage** without overwhelming the user?
- How can errors be **detected and corrected easily**?
- How can the interface support users with different levels of technical expertise?
- How can attendance information be presented through **effective visualizations**?
- How can the system maintain **consistency and visibility of system status**?

Project: Smart Student Attendance and Monitoring System – A Goal-Directed HMI Design

Objective:

Design and prototype a user-centered attendance management system using the **Goal-Directed Design process**. should conduct user research, develop personas, identify user goals, define requirements, design interaction flows and wireframes, develop a high-fidelity prototype, and conduct usability testing with representative users.

Minimum deliverables:

1. User research
2. User personas
3. User goals and task analysis
4. User scenarios
5. Requirements
6. Information architecture
7. User/task flow
8. Low-fidelity wireframes
9. High-fidelity interactive prototype
10. Usability testing
11. Usability findings
12. Iterative design improvements
13. Final prototype

1. Research – Understand the Users

The first stage of Goal-Directed Design is **Research**.

The system considers three major users:

- **Faculty** – wants to record attendance quickly and accurately.
- **Student** – wants to know their current attendance and shortage status.
- **Administrator** – wants to monitor attendance across classes and generate reports.

The **User Personas** section represents this stage.

For example:

> **Faculty goal:** “Mark attendance quickly and accurately.”
>
> **Student goal:** “Track my attendance and identify shortages.”
>
> **Administrator goal:** “Monitor attendance across all classes.”

The personas also identify **pain points**, which help the designer understand why the existing system needs improvement.

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2. Modeling – Convert Research into User Models

After research, GDD creates **personas and user models**.

The system shows three personas:

Persona	Goal	Pain Point
Faculty	Mark attendance quickly and accurately	Manual attendance takes time
Student	Track attendance and shortage	Does not know real-time status
Administrator	Monitor attendance across classes	Difficult to obtain consolidated reports

This is important because the designer does not design **one generic interface for everyone**.

Instead:

Faculty → Attendance interface

Student → Attendance/alert interface

Administrator → Monitoring dashboard

This is a key characteristic of **Goal-Directed Design**.

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3. Requirements Definition – What Does Each User Need?

The next stage converts user goals into **requirements**.

For example:

Faculty requirements

- View today's timetable
- Start attendance
- Identify
- Mark attendance
- Correct errors
- Review attendance
- Submit attendance
- Generate reports

Student requirements

- View overall attendance
- View subject-wise attendance
- Identify attendance shortage
- Receive alerts
- View attendance history

Administrator requirements

- View attendance statistics
- Compare divisions/classes
- Identify shortage
- Monitor trends
- Generate reports

The **Faculty Attendance Workflow** directly represents these requirements.

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4. Framework Definition – Design the Interaction Structure

This is one of the most visible parts of the design model.

The faculty workflow is:

Login → Today's Schedule → Start Attendance → Verify → Review & Correct → Submit → Summary

This is an example of designing the interface around the **user's goal and task sequence**.

Instead of giving faculty a complicated menu containing dozens of functions, the interface guides them through the actual task they want to accomplish:

“I want to record today's attendance.”

The **Today's Schedule** screen is particularly important.

The system already knows:

- Subject
- Class
- Time
- Number of

Therefore, the faculty does not need to repeatedly enter this information.

This demonstrates **Goal-Directed Design + reduced cognitive load**.

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5. Refinement – Detailed Interface Design

The next GDD stage is **Refinement**.

This is where the detailed UI shown in the design model is created.

Faculty Interface

The faculty dashboard contains:

- Today's Schedule
- Mark Attendance
- Student list
- Present/Absent/Late status
- QR/manual attendance options
- Search
- Review screen
- Confirmation

Notice the **Present / Absent / Late / Excused** summary.

This provides **visibility of system status**.

The faculty can immediately understand:

52 Present | 7 Absent | 2 Late | 1 Excused

rather than manually counting records.

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6. Goal-Directed Design and HMI Principles

The interface also demonstrates several important HMI principles.

Visibility of System Status

The system continuously shows attendance status.

For example:

52 Present

7 Absent

2 Late

The user does not have to guess whether the attendance has been recorded.

Recognition Rather Than Recall

The faculty does not need to remember:

- Which lecture is scheduled
- Which class is assigned
- How many are registered

The **Today's Schedule** automatically presents this information.

Similarly, the student interface displays subject-wise attendance directly.

Error Prevention

Before final submission, the system provides:

Review & Confirm → Confirm & Submit

This is important.

Instead of allowing the faculty to accidentally submit incorrect attendance, the system provides an opportunity to review it.

Feedback

After submission, the interface displays:

Attendance Submitted Successfully

with a confirmation indicator.

This provides immediate feedback that the user's action has been completed.

Consistency

The same concepts are represented consistently:

- Present
- Absent
- Late
- Excused
- Attendance percentage
- Alerts

Consistent visual representation makes the system easier to learn.

Error Recovery

The **Review & Correct** screen allows faculty to correct legitimate mistakes before final submission.

This supports the HMI principle:

> **Users should be able to recover from errors rather than being permanently affected by them.**

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7. Student Interface – Goal: “Know My Attendance”

The student interface is designed around a different goal.

The student does **not** need the complete administrative attendance database.

The student's primary question is:

> **“What is my attendance status?”**

Therefore, the interface provides:

Overall Attendance → Subject-wise Attendance → Attendance Details → Alerts

For example:

Overall: 78%

and subject-wise values such as:

- HCI – 82%
- Database Systems – 68%
- Software Engineering – 72%

The system also identifies **shortage subjects**.

This is goal-directed because the interface prioritizes the information the student actually needs.

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8. Administrator Interface – Goal: “Monitor Attendance”

The administrator has a completely different goal.

Instead of looking at one student's attendance, the administrator needs a **macro-level view**.

Therefore, the interface provides:

- Total
- Average attendance
- below threshold
- Total classes
- Attendance trends
- Department-wise attendance
- Shortage subjects
- Reports

This is why the design model contains an **Administrator Dashboard**.

The dashboard transforms large amounts of attendance data into **quickly understandable visual information**.

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9. Validation – Test Whether the Design Achieves the Goals

The final GDD stage is **Validation**.

developing this HMI project should test the prototype with representative users.

For example, give the faculty this task:

> **Task:** Record attendance for today's HCI lecture and submit it.

Measure:

- Time taken
- Number of clicks/interactions
- Errors made
- Whether the user understands the interface
- Whether the user can correct an error
- Whether the user knows that attendance was successfully submitted

For :

> **Task:** Find your attendance percentage in Database Systems and determine whether you have an attendance shortage.

For administrators:

> **Task:** Identify subjects where attendance is below the required threshold and generate a report.

The results should then be used to **modify the interface**.

Complete GDD Mapping of the design model

GDD Stage	What is Represented in the design model
1. Research	Understanding faculty, student and administrator problems
2. Modeling	Three user personas
3. Requirements Definition	User goals, workflows and required functions
4. Framework Definition	Navigation, attendance workflow and information architecture
5. Refinement	Detailed faculty, student and administrator interfaces
6. Validation	Usability testing and iterative improvements
7. Final Design	Final interactive attendance management interface

The key concept to teach

The most important distinction is:

Feature-driven design:

> “The system should have QR attendance, reports, dashboards, alerts, etc.”

Goal-directed design:

> “The faculty's goal is to record attendance quickly and accurately. Therefore, what is the minimum interaction needed to achieve that goal?”

Similarly:

> **Student goal:** Know attendance status → simple attendance dashboard.
>
> **Faculty goal:** Record attendance → schedule-driven attendance workflow.
>
> **Administrator goal:** Monitor attendance → analytical dashboard.

So, the design model demonstrates that **the interface is a consequence of user goals**, which is the central idea of **Goal-Directed Design**.

1. Universal Design Applied to the Attendance System

The design model is divided into seven sections. The first section shows the major Universal Design principles being evaluated.

A. Equitable Use

The system should provide comparable functionality to different users.

For example:

- Faculty can mark attendance.
- can view attendance.
- Administrators can monitor attendance.
- Users with accessibility requirements should also be able to perform their tasks.

Evaluation question:

> Can different users accomplish their intended task without being unnecessarily excluded?

B. Flexibility in Use

The system should support different ways of performing a task.

For example, the design model shows:

QR/automated attendance + manual attendance

A faculty member could use automated identification when available, but should also have an alternative method when it is not.

Similarly, may access the system through different devices.

Evaluation question:

> Does the interface provide appropriate alternatives when users have different preferences or circumstances?

C. Simple and Intuitive Use

The interface should not require extensive training.

For example, the student dashboard immediately shows:

Overall Attendance: 78%

and provides options such as:

- View Schedule
- View Attendance

- View Alerts
- View Details

The user should understand what to do without reading a lengthy manual.

Evaluation question:

> Can a first-time user understand the interface and complete the task?

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2. Perceptible Information

Information should be understandable through different forms of presentation.

For attendance, the system should not depend only on colour.

For example:

❌ Red = Absent

is insufficient by itself.

Instead:

🔴 **Absent**

or

Absent – 7

should be displayed.

This is particularly important for users with **colour-vision deficiencies**.

The design model also demonstrates:

- Icons
- Text labels
- Charts
- Status indicators
- Visual feedback

Evaluation question

> Can users understand important information even if they cannot perceive one particular visual characteristic?

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3. Tolerance for Error

A good Universal Design should help users **avoid errors and recover from them**.

For example, when faculty submits attendance, the system provides:

Review → Confirm & Submit

rather than immediately submitting the record.

The interface can also provide:

> "Are you sure you want to submit attendance?"

with:

Yes, Submit | Cancel

This reduces accidental submissions.

Evaluation question

> Does the interface prevent serious errors and provide an easy way to recover from mistakes?

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4. Low Physical Effort

The system should minimize unnecessary interaction.

For example, faculty should not have to:

1. Select department
2. Select semester
3. Select division
4. Select subject
5. Enter lecture time
6. Enter student list

every time attendance is recorded.

Instead, the system can use the timetable:

Today's Schedule → HCI → TE Comp A → Start Attendance

This reduces interaction effort.

Evaluation question

> Can users complete frequent tasks with minimum unnecessary interaction?

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5. Size and Space for Interaction

Buttons, controls, menus, and touch targets should have adequate size and spacing.

For example:

Confirm & Submit

should be clearly distinguishable from:

Cancel

This reduces accidental selection, particularly on touch devices.

The same principle applies to:

- Navigation buttons
- Attendance controls
- Search fields
- Filters
- Alerts

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6. Accessibility Considerations

The design model identifies five important accessibility areas.

Visual Accessibility

The interface should support:

- Adequate contrast
- Scalable text
- Clear typography
- Colour-independent status indicators
- Zoom

For example:

A− | A | A+

allows users to adjust text size.

Hearing Accessibility

If the system provides audio notifications, equivalent visual feedback should be available.

For example:

Instead of only:

🔊 *"Attendance submitted."*

provide:

✅ **Attendance submitted successfully**

Motor Accessibility

Users should not be forced to perform precise or repetitive actions.

The system can support:

- Keyboard navigation
- Larger controls
- Logical tab order
- Minimal steps

Cognitive Accessibility

The interface should reduce cognitive load through:

- Simple language
- Consistent navigation
- Clear headings
- Step-by-step workflows
- Familiar icons
- Clear error messages

For example:

Instead of:

> **ERR-ATD-503**

use:

> **Attendance could not be submitted. Please check your connection and try again.**

Language Accessibility

The design model demonstrates the possibility of **multi-language support**.

For example:

Attendance

could potentially be presented in:

- English
- Hindi
- Marathi

This can help users who are more comfortable with different languages.

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7. Evaluation Techniques

The design model also identifies several ways in which Universal Design can actually be **evaluated**.

A. Heuristic Evaluation

An HMI evaluator examines the interface against established usability and accessibility principles.

For example:

Evaluation Question	Observation
Is system status visible?	Attendance percentage is continuously visible
Is feedback provided?	Submission confirmation is displayed
Can users recover from errors?	Review & Correct option is available
Is information understandable?	Text + icons + charts are used
Is the interface consistent?	Similar controls use consistent layouts

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B. User Testing with Diverse Users

Instead of testing only with technically skilled , test with different categories of users.

The design model identifies:

- Faculty
-
- Administrators
- Elderly users
- Users with visual impairment
- Users with motor impairment
- Users in bright light
- Users with low digital literacy
- Users with slow Internet

This is an important Universal Design concept.

Example

Give the same task to different users:

"Check your attendance in HCI and determine whether you have a shortage."

Observe:

- Can they find the information?
- How long does it take?
- Do they make errors?
- Do they understand the percentage?
- Do they understand the shortage alert?

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8. Cognitive Walkthrough

The evaluator can perform the task step-by-step from the user's perspective.

For example:

Task: View attendance

Step 1: Login

Step 2: Open Dashboard

Step 3: Select Attendance

Step 4: Select Subject

Step 5: View percentage

At every step ask:

Will the user know what to do?

Will the user notice the correct control?

Will the user understand the feedback?

This is particularly useful for evaluating the interface for **first-time users**.

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9. Accessibility Checklist

The interface can also be evaluated using an accessibility checklist.

For example:

Check	Yes/No
Sufficient colour contrast	✓

Information not conveyed by colour alone	✓
Text can be enlarged	✓
Keyboard navigation available	✓
Buttons adequately sized	✓
Clear error messages	✓
Consistent navigation	✓
Alternative text for important icons/design models	✓
Accessible forms	✓
Clear labels	✓

For a formal academic project, can relate this evaluation to **WCAG** accessibility guidelines.

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10. Task Success Rate

The final evaluation should measure whether users can actually accomplish their goals.

For example:

Faculty Task

> Mark attendance for TE Computer Engineering – Division A.

Measure:

- Completion rate
- Time taken
- Number of errors
- Number of interactions
- User satisfaction

Student Task

> Find the attendance percentage for Database Systems.

Measure:

- Task completion
- Time taken
- Errors
- User understanding

Administrator Task

> Identify subjects with attendance below the required threshold.

Measure:

- Task completion
- Information-finding time
- Errors
- Satisfaction

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11. What the Evaluation Should Ultimately Establish

The bottom section of the design model shows the desired outcomes:

Increased Usability → Higher Accessibility → Reduced Errors → Greater Satisfaction → Compliance Readiness

The ultimate goal is:

> **An attendance system that can be used effectively by the maximum number of users without requiring a separate or specialized interface.**

How to Explain -

Traditional usability evaluation

> **"Can a normal student use the attendance system easily?"**

Universal Design evaluation

> **"Can with different ages, abilities, languages, levels of digital literacy, devices, and environmental conditions use the attendance system effectively?"**

That difference is the **core concept of Universal Design as an evaluation technique**.

Simple evaluation framework

Users → Tasks → Universal Design Principles → Evaluation → Problems → Redesign → Re-evaluation

For this project:

Faculty / Student / Administrator
↓
Mark / View / Monitor Attendance
↓
Equitable + Flexible + Simple + Perceptible + Error-tolerant + Accessible
↓

Heuristic Evaluation + Cognitive Walkthrough + Diverse User Testing + Accessibility Checklist

↓

Identify Usability Problems

↓

Redesign Interface

↓

Test Again